

Table 5. **The output contract: 93 works**, selected by a stated rule: every work whose confidence is attached to a typed decision (contract “typed”, 65, at any stage), and the X-stage works on grammar- or schema-constrained output (28). Cal.: ✓ when the main text reports a calibration metric (ECE, Brier score, NLL or a reliability diagram); 41 of the 65 typed works do, all 7 in the training core. The constrained-output works mostly measure validity, speed or accuracy under the constraint. Each work’s contribution is in Table 3 or Table C1. Yr: year of the first public version; Family: the training family (S4) or method family; Metric: the calibration or confidence metric reported in the main text (ECE, Brier, NLL, AUROC, selective = selective risk or accuracy at a coverage, coverage = achieved coverage of a prediction set or achieved risk under a controlled threshold, multiple = more than one of these (a reliability diagram counts with ECE; PRR with AUROC), other = a different metric, none = no confidence metric). ? = not determinable from the paper.

Work	Yr	Family	Metric	Cal.
Typed output, S0 read-off (16)				
OpenAI [2023]	2023	Calibration after alignment	ECE	✓
Huang et al. [2026]	2026	Calibration after alignment	ECE	✓
Proskurina et al. [2026]	2026	Calibration after alignment	ECE	✓
Sanz-Guerrero et al. [2026]	2026	Calibration after alignment	multiple	✓
Desai and Durrett [2020]	2020	Empirical studies of read-off calibration	ECE	✓
Dan and Roth [2021]	2021	Empirical studies of read-off calibration	ECE	✓
Ahuja et al. [2022]	2022	Empirical studies of read-off calibration	ECE	✓
Chen et al. [2023a]	2022	Empirical studies of read-off calibration	ECE	✓
Xiao et al. [2022]	2022	Empirical studies of read-off calibration	multiple	✓
Li et al. [2025h]	2023	Empirical studies of read-off calibration	multiple	✓
Zhang et al. [2023d]	2023	Empirical studies of read-off calibration	ECE	✓
Mahaut et al. [2024]	2024	Empirical studies of read-off calibration	other	–
Joshi et al. [2025]	2025	Empirical studies of read-off calibration	ECE	✓
Lovering et al. [2025]	2024	Likelihood and logit read-off	other	–
Plaut et al. [2024]	2024	Likelihood and logit read-off	multiple	✓
Wang et al. [2024d]	2024	Likelihood and logit read-off	none	–
Typed output, S1 post-hoc map (24)				
Badhe et al. [2026]	2026	Fixed maps over the read-off	multiple	✓
Zheng et al. [2023]	2023	Label-bias and option-prior debiasing	none	–
Reif and Schwartz [2024]	2024	Label-bias and option-prior debiasing	none	–
Han et al. [2022]	2022	Learned calibrators	none	–
Luo et al. [2026]	2026	Learned calibrators	multiple	✓
Liu et al. [2024b]	2024	Multicalibration and task-specific recalibration	ECE	✓
Burns et al. [2022]	2022	Probes and learned confidence heads	other	–
Azaria and Mitchell [2023]	2023	Probes and learned confidence heads	AUROC	–
Liu et al. [2023a]	2023	Probes and learned confidence heads	ECE	✓
Holtzman et al. [2021]	2021	Prompt and context calibration	none	–
Zhao et al. [2021]	2021	Prompt and context calibration	none	–
Fei et al. [2023]	2023	Prompt and context calibration	none	–
Jiang et al. [2023b]	2023	Prompt and context calibration	none	–
Liusie et al. [2023]	2023	Prompt and context calibration	none	–
Zhou et al. [2023b]	2023	Prompt and context calibration	none	–
Abbas et al. [2024]	2024	Prompt and context calibration	ECE	✓
Cho et al. [2025]	2024	Prompt and context calibration	none	–
Gundem et al. [2025]	2025	Prompt and context calibration	none	–
Guo et al. [2017]	2017	Scaling and binning	multiple	✓
Kull et al. [2019]	2019	Scaling and binning	multiple	✓
Kumar et al. [2019]	2019	Scaling and binning	multiple	✓
Gupta and Ramdas [2021]	2021	Scaling and binning	ECE	✓
Shen et al. [2024]	2024	Scaling and binning	multiple	✓

Table 5, continued

Work	Yr	Family	Metric	Cal.
Luo et al. [2025]	2025	Scaling and binning	multiple	✓
Typed output, S3 extra inference (11)				
Ling et al. [2024]	2024	Benchmarks and empirical studies	AUROC	–
Jung et al. [2024]	2024	Conformal abstention, selection and cascades	multiple	✓
Tonolini et al. [2024]	2024	Ensembles and cross-model disagreement	multiple	✓
Kruse et al. [2025]	2025	Ensembles and cross-model disagreement	multiple	✓
Schuster et al. [2021]	2021	Risk control and early exit	coverage	–
Jazbec et al. [2024]	2024	Risk control and early exit	multiple	✓
Liu et al. [2023b]	2023	Sampling consistency	other	–
Portillo Wightman et al. [2023]	2023	Sampling consistency	ECE	✓
Becker and Soatto [2024]	2024	Sampling consistency	multiple	✓
Yang et al. [2024a]	2024	Sampling consistency	multiple	✓
Zhang et al. [2024a]	2024	Sampling consistency	ECE	✓
Typed output, S4 training (7)				
Yaldiz et al. [2026]	2026	T3 calibration rewards on the stated confidence	ECE	✓
Kong et al. [2020]	2020	T1 supervised calibration objectives	ECE	✓
Niu et al. [2024]	2024	T1 supervised calibration objectives	ECE	✓
Wang et al. [2024e]	2024	T1 supervised calibration objectives	multiple	✓
Xiao et al. [2025]	2025	T1 supervised calibration objectives	ECE	✓
Xie et al. [2026]	2026	T1 supervised calibration objectives	multiple	✓
Jiang et al. [2021]	2020	T1 supervised calibration objectives	ECE	✓
Typed output, X consumers and contract (7)				
Jitkrittum et al. [2023]	2023	Deferral	none	–
Kondadadi and Ortega [2026]	2026	Deferral	none	–
Li et al. [2026c]	2026	Judge calibration	multiple	✓
Varshney and Baral [2022]	2022	Routing/cascade	none	–
Ramirez et al. [2024]	2024	Routing/cascade	none	–
Zellinger and Thomson [2025]	2025	Routing/cascade	ECE	✓
Wang et al. [2024c]	2024	Typed classification	none	–
Constrained output, X consumers and contract (28)				
Scholak et al. [2021]	2021	Constrained decoding	none	
Beurer-Kellner et al. [2023]	2022	Constrained decoding	none	
Poesia et al. [2022]	2022	Constrained decoding	none	
Geng et al. [2023]	2023	Constrained decoding	none	
Lew et al. [2023]	2023	Constrained decoding	none	
Willard and Louf [2023]	2023	Constrained decoding	none	
Beurer-Kellner et al. [2024]	2024	Constrained decoding	none	
Dong et al. [2025]	2024	Constrained decoding	none	
Geng et al. [2024b]	2024	Constrained decoding	none	
Koo et al. [2024]	2024	Constrained decoding	none	
Park et al. [2024]	2024	Constrained decoding	none	
Ugare et al. [2024]	2024	Constrained decoding	none	
Banerjee et al. [2025]	2025	Constrained decoding	none	
Li et al. [2026b]	2025	Constrained decoding	none	
Loula et al. [2025]	2025	Constrained decoding	other	
Mündler et al. [2025]	2025	Constrained decoding	none	
Park et al. [2025b]	2025	Constrained decoding	none	
Reddy et al. [2026]	2026	Constrained decoding	none	
Tang et al. [2023]	2023	Structured-output benchmark	none	
Shorten et al. [2024]	2024	Structured-output benchmark	none	
Tam et al. [2024]	2024	Structured-output benchmark	none	
Xia et al. [2024a]	2024	Structured-output benchmark	none	

Table 5, continued

Work	Yr	Family	Metric	Cal.
Geng et al. [2025]	2025	Structured-output benchmark	none	
Yang et al. [2025a]	2025	Structured-output benchmark	none	
Fan [2026]	2026	Structured-output benchmark	none	
Galeone et al. [2026]	2026	Structured-output benchmark	none	
Lee et al. [2026a]	2026	Structured-output benchmark	none	
Singh et al. [2026a]	2026	Structured-output benchmark	none	